Business Problem

Customer retention is critical for sustainable growth in the highly competitive OTT streaming industry. Businesses may lose subscribers and revenue due to subscription-related issues, customer dissatisfaction, and poor support experiences. The goal of this analysis is to:

- Evaluate customer churn and retention trends across time, subscription plans, and states.
- Assess customer revenue, subscription tenure, and potential revenue loss due to churn.
- Identify patterns in churn risk across customer and subscription attributes.
- Examine customer support escalations and complaints in relation to churn.
- Compare monthly charges across customer groups to identify useful retention insights.

Data Analysis Insights

Research Questions & Key Findings

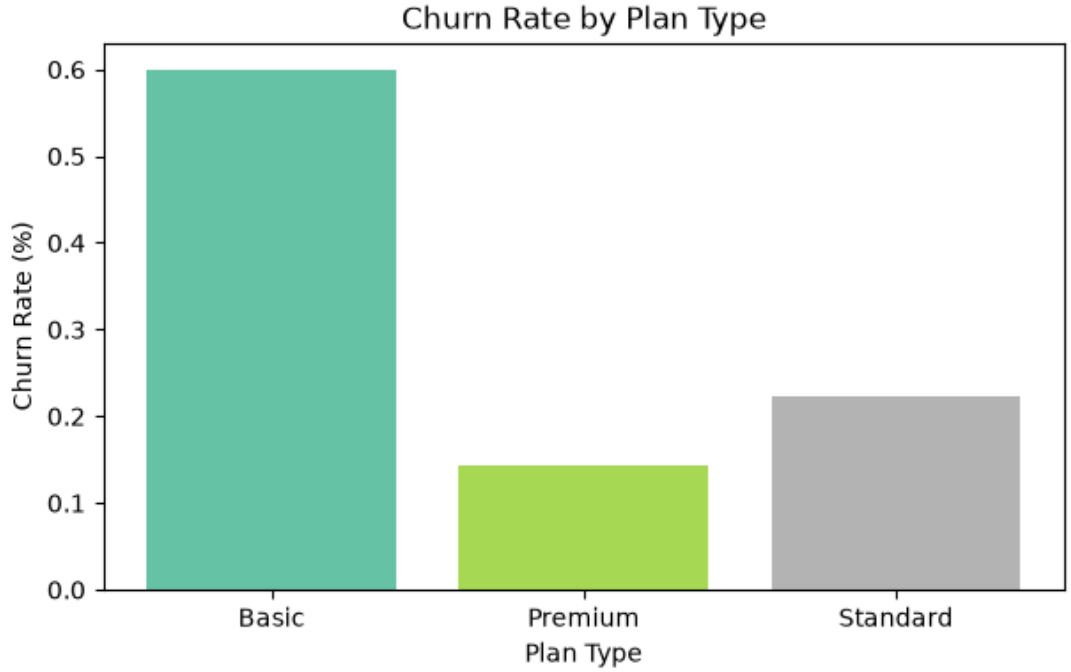
1. Customer Churn and Retention Rate Analysis

- Customer Churn Rate: The churn rate stands at 28.6%, highlighting the share of customers lost through subscription cancellations.
- Customer Retention Rate: With a retention rate of 71.4%, the business has retained more than two-thirds of its customers, providing a useful basis for evaluating customer stability.

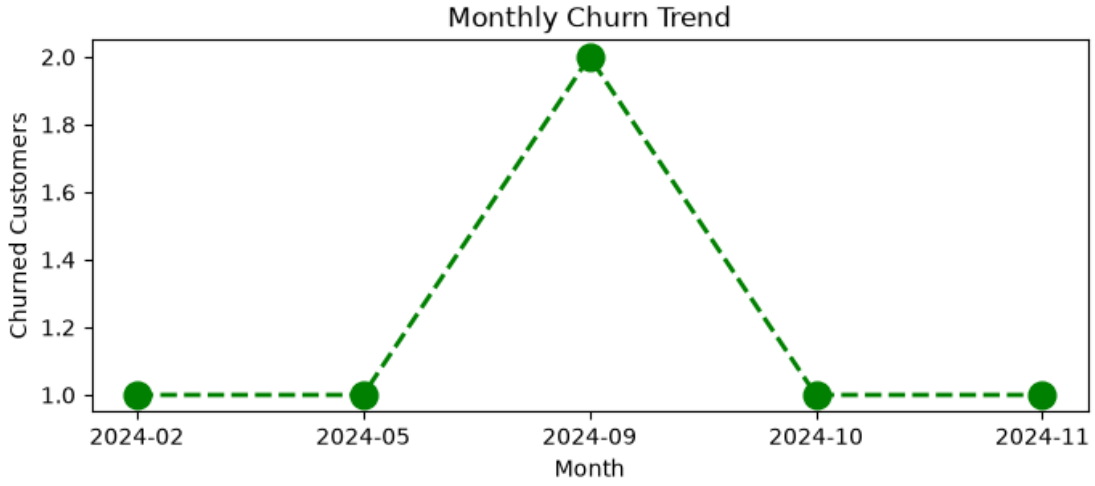
2. Customer Churn Analysis by Subscription Plan

	plan_type	churn_rate_pct
0	Basic	60.00
1	Premium	14.29
2	Standard	22.22

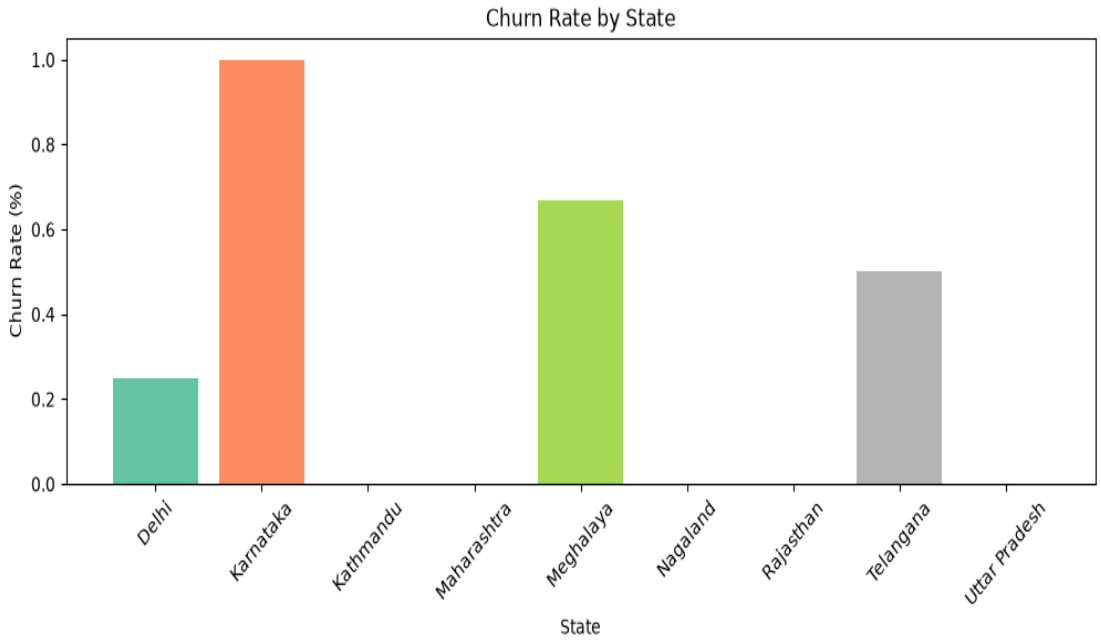
The Basic plan has the highest churn rate at 60.00%, compared with 22.22% for Standard and 14.29% for Premium. The Premium plan shows the lowest churn rate, while the Basic plan contributes the largest share of customer cancellations relative to its subscribers.



3. Monthly and State-Wise Customer Churn Analysis



September 2024 experienced the highest customer churn, with 2.0 cancellations, while the other recorded months had 1.0 cancellation each. This indicates a relatively consistent pattern of cancellations, with a slight increase during September.



Customer churn patterns vary across states, with Karnataka recording the highest churn, followed by Meghalaya and Telangana. Delhi experienced comparatively lower churn, showing differences in subscription cancellation trends across geographical locations.

4. Average Customer Tenure and Revenue Per User

- Average Customer Tenure: The average customer tenure is 1,451 days, showing the length of time customers have used the service.
- Average Revenue Per User: The ARPU is 18.8, providing a measure of the revenue generated per customer through monthly charges.

5. Revenue Loss Due to Customer Churn

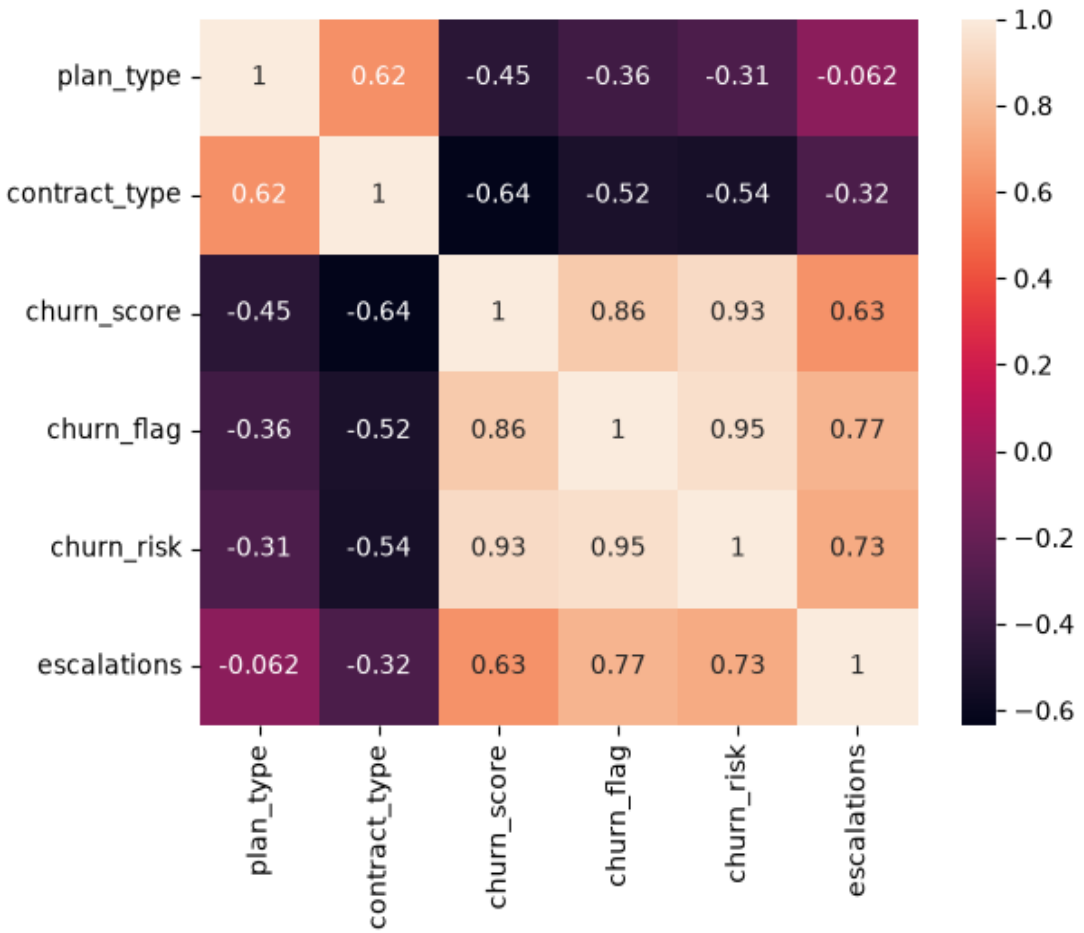
Revenue associated with churned customers is estimated at 74, based on their monthly charges. This represents the recurring revenue no

longer expected from customers who discontinued their subscriptions, demonstrating the financial effect of customer cancellations.

6. Percentage of Revenue Loss Due to Churn

Customer cancellations led to an 18% loss in total revenue, reflecting the financial effect of subscribers leaving the service. The remaining revenue provides a foundation for assessing the financial impact of churn and monitoring revenue stability.

Correlation Insights



Monthly Churn vs. Annual Churn: Monthly subscriptions have a churn rate of 55.6%, compared with 8.3% for annual subscriptions. Annual subscribers show a lower tendency to cancel than monthly subscribers.

Final Recommendations

- Re-evaluate the reasons behind the high churn rate among Basic plan subscribers and check whether pricing or service issues contribute to cancellations.
- Enhance customer retention by prioritizing customers with medium and high churn risk and addressing their concerns before they cancel their subscriptions.
- Re-evaluate the difference in churn rates between monthly and annual subscribers to identify ways to improve subscription retention.
- Investigate customer feedback and service experiences in states with higher churn rates, particularly Karnataka.
- Optimize revenue monitoring by tracking the financial impact of customer cancellations and using these insights to support retention decisions.
- The successful implementation of these recommendations can help the company achieve sustainable profitability, minimize business risks, and strengthen overall operational efficiency.